MTH636 – Game Theory

Lecture 8: Iterated Elimination & Nash Equilibria

1 Iterated Elimination of Strictly Dominated Strategies (IESDS)

Recall that for a game $G = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$, a strategy profile s^* is a Nash Equilibrium if $u_i(s^*) \geq u_i(s_i, s^*_{-i})$ for all $i \in N, s_i \in S_i$.

Suppose we create a new game $\hat{G} = \langle N, (S'_i)_{i\in N}, (u_i)_{i\in N}\rangle$ by removing a **strictly dominated strategy** $\bar{s}_k$ for some player k. So:

$$S'_i = S_i \text{ for } i \neq k, \quad \text{and} \quad S'_k = S_k \setminus \{\bar{s}_k\}$$

Theorem: IESDS and Nash Equilibria

The set of Nash Equilibria in G and in $\hat{G}$ remain **exactly the same**.

1. **IESDS cannot remove any Nash Equilibrium:** If an NE were removed, it must have had $\bar{s}_k$ as its k-th component. But a strictly dominated strategy can never be a best response, so it can never be part of a Nash Equilibrium.
2. **IESDS cannot create new Nash Equilibria:** Could a profile s^* be an NE in $\hat{G}$ but not in G? If it wasn't an NE in G, some player must have had a profitable deviation. Since all strategies except $\bar{s}_k$ are still in $\hat{G}$, the only way the deviation is missing is if the profitable deviation was to $\bar{s}_k$ itself! But $\bar{s}_k$ is strictly dominated by some $t_k \in S'_k$, meaning t_k would be an even better deviation that is still available in $\hat{G}$. Thus, no new NE is created.

2 Iterated Elimination of Weakly Dominated Strategies

The strong guarantee of IESDS **does not hold** for weakly dominated strategies.

Key Insight: Weakly Dominated Strategies can remove NE

Consider the following game:

	L	R
T	(0, 0)	(2, 1)
B	(3, 2)	(1, 2)

Nash Equilibria of Original Game: Both (B, L) and (T, R) are Nash Equilibria.
Elimination:

- For Player 2, R weakly dominates L (payoffs: $1 \geq 0$ and $2 \geq 2$).
- If we eliminate L, we are left with the right column.
- In the reduced game (T vs R yields 2, B vs R yields 1), T strictly dominates B for Player 1. We eliminate B.
- We are left with only (T, R).

By eliminating weakly dominated strategies, we **lost** the Nash Equilibrium (B, L)!

Question: Can iterated elimination of weakly dominated strategies create *new* Nash Equilibria that were not NE in the original game? (No, but proving it requires care).

3 Introduction to Mixed Strategies

Until now, we assumed players choose a specific strategy deterministically (Pure Strategies). But what if players can randomize?

Consider a choice between:

- Option A: Receive Rs 100 for sure.
- Option B: Receive Rs 500 with probability 1/4, and Rs 0 with probability 3/4.

We assume players are **rational** and evaluate probability distributions using **Expected Utility**. The expected value of B is $\frac{1}{4}(500) + \frac{3}{4}(0) = 125$. If utility is linear in money, the player prefers B over A.

3.1 Affine Transformations of Utilities

Suppose we have a game G with payoffs u_i. We create an equivalent game $\hat{G}$ by applying a function to the utilities: $\hat{u}_i = v_i \circ u_i$.

- If v_i is a general strictly increasing monotonic transformation, it preserves preferences over **pure** strategies, but it **does not** preserve preferences over **mixed** (randomized) strategies, because it changes the expected values relative to each other.
- To preserve preferences over mixed strategies, we must only allow **Positive Affine Transformations**:
$$v_i(x) = a_i x + b_i \quad \text{where } a_i > 0$$